Understanding Genetic Traits: From Simple to Complex

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```
try:
    import au_molecular_genetics
    print("Already installed")
except ImportError:
    %pip install -q "au_molecular_genetics @ git+https://github.com/au-mbg/molecular_
genetics.git"
```

```
from au_molecular_genetics.trait_simulation import
generate_mendelian_quantitative_comparison
```

This notebook helps us visualize how different numbers of genes contribute to a trait, and how the environment can influence it. We're simulating traits in a population of $N = 500$ individuals.

Key Concepts:

- **Mendelian Traits (1 gene)**: These are traits determined by a single gene. Think of simple inheritance patterns like Mendel's pea plants. In our simulation, `n_loci=1` represents this.
- **Polygenic Traits (Multiple genes)**: These traits are influenced by many genes, each contributing a small effect. Most complex human traits, like height or intelligence, are polygenic. Our simulations with `n_loci=2` and `n_loci=3` model increasingly polygenic traits.
- **Alleles and Genetic Score:** Each gene has different versions called alleles. We're tracking a 'plus' (+) allele which contributes to the trait. Your 'genetic score' is simply the total number of these '+' alleles you have across all the genes we're considering.
- **Environmental Noise (env_sd)**: In real life, traits aren't just about genes; the environment also plays a role. `env_sd` represents how much random environmental variation is added to the genetic effect. A low `env_sd` means the environment has little impact, while a high `env_sd` means it has a significant impact.

What the Plots Show:

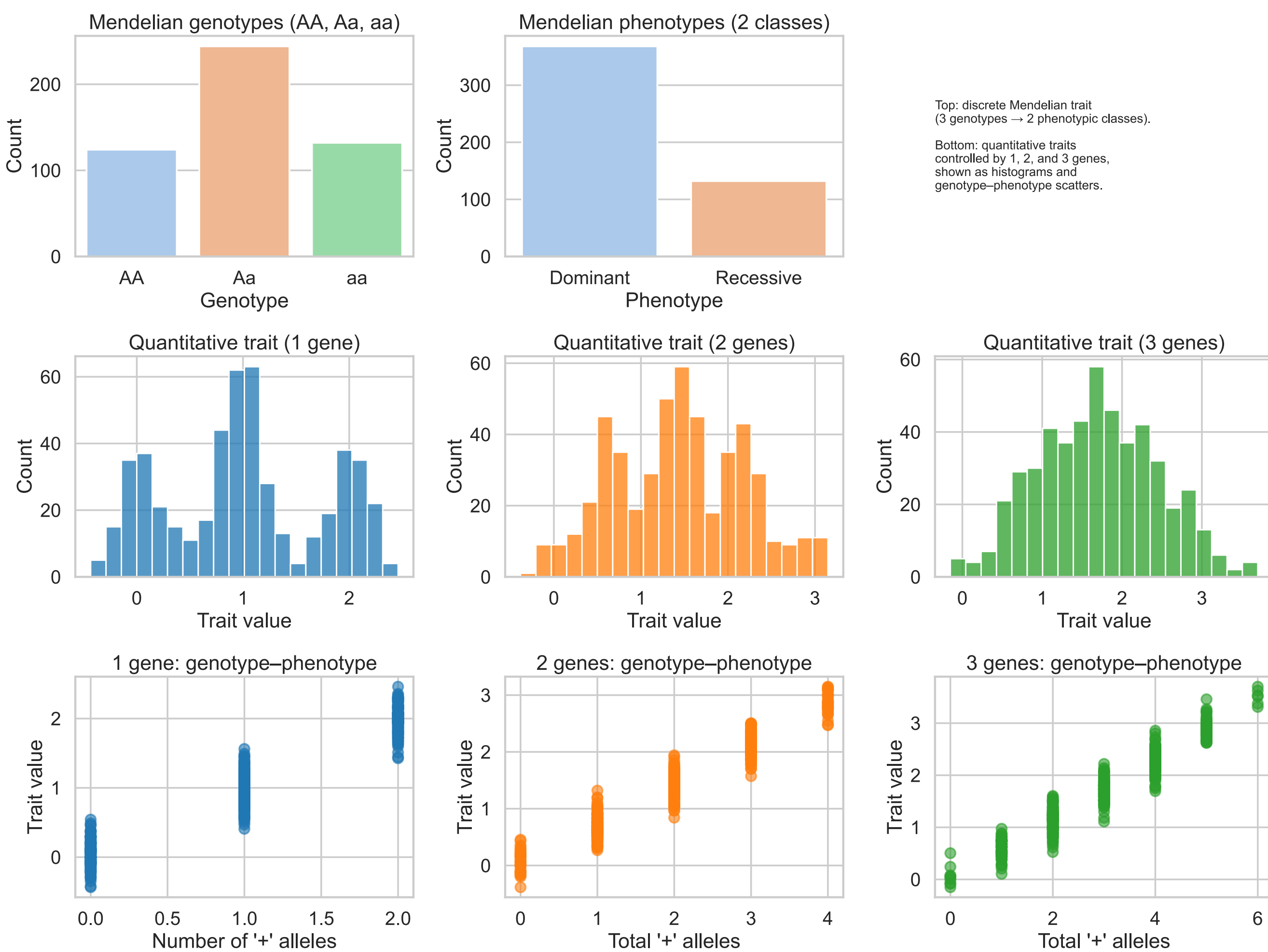
- **Individual vs. Trait**: This scatter plot shows the trait value for each of our 500 simulated individuals. It gives you a raw look at the variation in the trait across the population.
- **Trait Distributions**: These histograms show the overall shape of the trait values in the population. For single-gene traits with no environmental noise, you'll see distinct 'classes' of phenotypes. As more genes are added, or environmental noise increases, the distribution tends to become more continuous and bell-shaped.
- **Genetic Score vs. Trait**: This crucial plot shows how your genetic score (number of '+' alleles) relates to your actual trait value. For Mendelian traits with no environmental noise, there's a perfect, clear relationship. As environment comes into play, this relationship becomes fuzzier.

Run the cell below to generate the plots.

```python
# Core parameters
N = 500
p = 0.5

# Environmental SD per gene category
env_sd1 = 0.2
env_sd2 = 0.2
env_sd3 = 0.2
n_loci = (1, 2, 3)

# Run the simulation
generate_mendelian_quantitative_comparison(N=N, p=p, env_sd1=env_sd1, env_sd2=env_sd2,
env_sd3=env_sd3)
```




```
Mendelian phenotypic class counts:
Phenotype
Dominant     368
Recessive    132
Ratio Dominant:Recessive = 368:132 (≈ 2.79:1)
```

```
Quantitative means: 1.0082470180067222 1.4331142971547959 1.6988601109688748
Quantitative SDs: 0.7361215533808997 0.7504735313164385 0.7478658519882052
```

1 Your Turn:

Looking at the 1-gene trait (Mendelian) and the summary table of its phenotypic classes, how do you think changing the sample size (N) from 500 to, say, 50 or 5000 would affect the sampling error in the observed ratios of these phenotypic classes? What might you expect to see if you ran the simulation with a much smaller or much larger number of individuals?